Project Completion Guide

Monitoring Deforestation and Land-Use Change in the Amazon: from the finished deck to real results

1. Where the project stands

Done: the 21-slide deck (PowerPoint and PDF) and the six code files in this package.

Not done, and only you can do it: no model has been trained yet. My workspace had no GPU, no PyTorch and no access to Kaggle, so I could not download the data or train anything. That means:

- Slide 14 (the scorecard) is empty on purpose. Every number in it must come from your own run.
- All satellite-style pictures in the deck are synthetic illustrations. Script 03 makes real ones.
- The data-prep and evaluation code (scripts 02, 03 and 05) was tested on fake data and runs. The training and Grad-CAM code (scripts 04 and 06) was only syntax-checked, so expect to fix small things the first time you run it. Send me any error message and I will fix the code.

Time needed (estimates, they vary with Colab): about 3 to 5 hours of mostly waiting. Download and extraction 30 to 90 minutes, data prep 10 to 20 minutes, training 2 to 3 hours for all four models on a free T4 GPU, evaluation and figures 10 minutes, then 1 to 2 hours to update the deck.

2. Files in this package

File	What it does	GPU?
01_setup_download	Installs tools, connects Kaggle, downloads and unpacks the data	No
02_eda_and_cache	Label counts, band-order check, NDVI, 128 px cache, train/val/test split	No
03_real_figures	Real image panels to replace the synthetic ones in the deck	No
04_train_models	Trains experiments A, B, C, D	**Yes**
05_evaluate	Thresholds, test metrics, significance test, curves, confusion matrices	No
06_gradcam_and_errors	Grad-CAM heatmaps and error analysis by weather	Helpful

Experiments: **A** scratch CNN with RGB, **B** scratch CNN with RGB+NIR+NDVI, **C** pretrained ResNet-34 with RGB, **D** pretrained ResNet-34 with RGB+NIR+NDVI.

3. Before you start (10 minutes)

1. Create or log in to a free **Kaggle** account.
2. Open the competition page *Planet: Understanding the Amazon from Space* and click the button to join or accept the rules. Without this the download fails with a 403 error.
3. In Kaggle, go to Settings, then API, and create a token. Keep the `kaggle.json` file it gives you (if Kaggle shows a token string instead, follow the instructions on that page).
4. Open **Google Colab** (colab.research.google.com), create a new notebook and choose Runtime, Change runtime type, **T4 GPU**.
5. Make sure your Google Drive has about 8 GB free. The scripts save the cache and results to a Drive folder called `amazon_project` so nothing is lost if Colab disconnects.

How to run each script: copy it into Colab. Every `# %%` line marks a cell, so paste each block into its own cell and run the cells in order. Run the scripts in the order 01, 02, 03, 04, 05, 06. If Colab asks to restart the runtime after installing packages, do it and continue.

4. Step-by-step

Step 1. Script 01: download

- Run all cells. Upload `kaggle.json` when asked.
- Cell 4 prints the competition's file list. If the file names differ from `train_v2.csv` and `train-tif-v2.tar.7z`, edit the FILES list in cell 5.
- The 4-band TIF files are required because only they contain the near-infrared band. The JPG files are RGB only and cannot give you NDVI.
- **Success check:** the last cell prints about 40,479 TIF files.
- **If disk space runs out:** delete the downloaded archive after extraction, or use Colab's larger disk, or run on Kaggle Notebooks where the dataset is already attached (you would change `paths.json`).

Step 2. Script 02: data prep

- Leave `SUBSET = None` for the full dataset. If Colab is slow or your GPU quota is short, set `SUBSET = 15000` (keeps every rare-label tile).
- **Check the band order.** The script prints the mean NDVI for primary forest, agriculture and water. Forest must be highest and water lowest. Open `figs/fig_band_check.png`: the top row (forest) must look green and the bottom row (cleared land) tan or brown. If the colours look wrong, tell me and I will help you fix the order.
- Outputs for the deck: `fig_label_counts.png`, `label_counts.csv` (the real class-imbalance numbers), `fig_label_cooccurrence.png`.

Step 3. Script 03: real images

- Run it once. Download the folder `figs/real` from Drive. You will use these images to replace the synthetic ones (section 6).

Step 4. Script 04: training (the long step)

- Needs the GPU runtime. Defaults: 14 epochs, batch 128, rotations and flips, oversampling of rare labels, class-weighted loss, mixed precision.
- It trains A, B, C, D one after another and saves each to `runs/`. **If Colab disconnects, reconnect, rerun cells 1 to 5 and then cell 6.** Finished experiments are skipped; the one that was running restarts.
- If you get a CUDA out-of-memory error, set `BATCH = 64`.

- If time is tight: reduce EPOCHS to 8 to 10, or use SUBSET in script 02. Say so on the slides.
- **Success check:** each epoch prints a validation macro F2 that rises over time; the `runs/` folder ends with four `_probs.npz` files.

Step 5. Script 05: evaluation

- Runs in seconds. It picks per-label thresholds on the **validation** set only and then measures everything on the **test** set once. Do not tune anything on the test results.
- Outputs: `results_table.csv`, `slide14_table.md` (ready to copy into slide 14), `bootstrap_differences.csv` (tells you whether NDVI and transfer learning gains are real or noise), `fig_results.png`, `fig_per_label_f2.png`, `fig_pr_rare.png`, `fig_confusion_matrices.png`.

Step 6. Script 06: interpretability

- Produces Grad-CAM images (`fig_gradcam_*.png`) for the rare labels, comparing the RGB-only and NIR+NDVI ResNets, and `fig_error_by_weather.png` showing whether clouds and haze cause the mistakes.

5. Report the metrics honestly (important)

- Your proposal's goal is **macro F2 of at least 0.85**. Macro F2 gives each of the 17 labels equal weight, and the rare ones (mining, logging, burning) are very hard, so macro F2 will probably land well below 0.85 even for a good model. The published Kaggle scores (0.93 and similar) use the **per-tile (samples) F2**, which script 05 prints as "Kaggle-style F2".
- Report both. Use the Kaggle-style F2 when you compare with the benchmark chart on slide 13, and macro F2 plus rare-class recall for your own success criteria.
- If you miss a target, say so on slide 12 and explain why. Judges respect an honest, well-explained result far more than a suspiciously perfect one.
- Always show the bootstrap result. "NDVI improved F2 by +0.0xx (95% interval … to …)" is a stronger claim than "NDVI seems better". If the interval includes zero, say the gain is not statistically clear.

6. Update the deck (slide by slide)

Open the .pptx in PowerPoint. To replace a picture: right-click, Change Picture, From a File. Then delete the small "illustrative" caption under it.

Slide	What to change
1 and 20	Replace the background image with `real_mosaic.png` (Insert, Picture, then Send to Back).
2	Check the bars for 2004, 2012 and 2021 against INPE's TerraBrasilis (terrabrasilis.dpi.inpe.br). Fix any value by right-clicking the chart, Edit Data.
3	Replace the picture with `real_chip_agriculture.png`.
4	Confirm the tile count from script 02. Replace the 8 chips with `real_chip_*.png` (primary, agriculture, road, water, partly_cloudy, haze, slash_burn, artisinal_mine). Delete the "illustrative" note.
5	Replace the picture with `real_multilabel.png` and **edit the four tag pills to match the tags printed by script 03**.
6	Put the real percentages from `label_counts.csv` in the imbalance card. Use `real_chip_artisinal_mine.png`, `real_chip_haze.png` and a road tile.

Slide	What to change
8	Use `real_rgb.png`, `real_cir.png`, `real_ndvi.png`. Check the descriptions still match.
9	Use `real_aug_0.png` to `real_aug_3.png`.
12	State whether each target was met, with your numbers.
13	Add a bar for your best model (Kaggle-style F2). Right-click the chart, Edit Data.
14	Fill the table from `slide14_table.md`. Delete the yellow "Fill in" label. Add one sentence on the bootstrap result.
15	Use `fig_gradcam_*.png` and `fig_error_by_weather.png` and write what you actually see.
16	The dataset has no repeat visits, so there is no real before/after pair. Either delete this slide, or keep it and keep its "illustrative" label, or replace it with a real before/after pair from NASA Earth Observatory or Planet with proper credit.
18	Edit the risks so they match what really happened (for example, if rare classes stayed hard, say so).
21	Update the line saying the images are synthetic if you replaced them all. Add the sources for your pictures if any are from outside.
All	Update the speaker notes where they say "synthetic" or "to be filled in".

You can also send me the output files (the CSVs and figures) and I will rebuild the deck for you.

7. If your results are weaker than hoped

- Train longer (`EPOCHS = 20`) or use larger tiles (`SIZE = 256` in script 02; slower).
- Try a different backbone (EfficientNet) or lower the ResNet learning rate to `5e-4`.
- Check the rare labels first: look at `fig_pr_rare.png` and the per-label F2 chart.
- Whatever happens, the 2x2 experiment is still a valid result. "NDVI did not help the pre-trained model, and here is why" is a complete scientific finding.

8. Likely judge questions

1. **Why F2 and not accuracy?** Missing real deforestation costs more than a false alarm, and accuracy hides the rare classes.
2. **How did you avoid data leakage?** Thresholds and the best epoch were chosen on validation data; the test set was used once.
3. **Is NDVI really helping?** Cite the bootstrap interval.
4. **Why 128 px?** Faster training; the cs231n project report found small inputs work well on this dataset. Say if you tested 256.
5. **Would it work on another year or region?** Not tested; name it as the next step (temporal and spatial hold-out).
6. **Can a model accuse someone of illegal logging?** No. Predictions are leads for human review.
7. **How do you know the model looks at the right thing?** Grad-CAM, with examples of good and bad cases.

9. Final checklist

☐ Kaggle rules accepted; `kaggle.json` uploaded
☐ Script 01 finished; about 40,479 TIFs found
☐ Script 02: band check looks right (green forest, NDVI highest for forest)
☐ Script 03 images downloaded

- ☐ Script 04: four `_probs.npz` files exist
- ☐ Script 05: results, bootstrap and figures saved
- ☐ Script 06: Grad-CAM and weather-error figures saved
- ☐ Slide 14 filled; yellow label removed
- ☐ Slides 12 and 13 reflect the real numbers
- ☐ Synthetic images replaced (or clearly labelled)
- ☐ Slide 2 figures checked against INPE
- ☐ Speaker notes and references updated
- ☐ Rehearsed once with a timer (about 1.5 to 2 minutes per slide)
- ☐ Exported the final PDF and saved a copy of the .pptx

10. Division of work (matches your proposal)

- **OluwaKoretimi Adebowale:** scripts 01 to 03 (data, NDVI, imbalance analysis).
- **Josselin Jimenez:** script 04 (models, hyperparameters, thresholds, tracking).
- **Mehrab U Azan:** scripts 05 and 06 (metrics, confusion matrices, Grad-CAM).
- **Everyone:** update the deck and rehearse together.